Probability

Approaches and Parameter Estimation

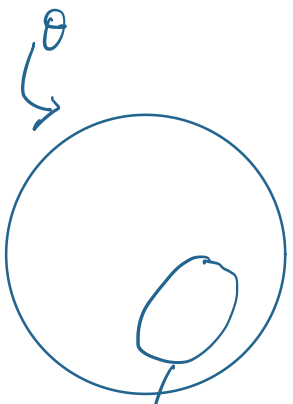
Frequentist vs Bayesian

- Frequentists
  - Treat parameters as fixed but unknown
  - Rely solely on observed data to make inferences

- Bayesian
  - Treat parameters as random variables with a defined prior
  - Updated beliefs with data to form a posterior distribution

এরকম অনেক model আছে যেটা থেকে θ appear করা probable

prior নিবো → data দেখবো → posterior update করবো



prior knowledge বলছে θ এর value এই region থেকে আসা উচিত

Frequentist vs Bayesian

e.g. bias coin কিনা
multiple times sample করে দেখবো

- Frequentists
  - Training data is a random draw from the population
  - Uncertainty in estimates is quantified through the sampling distribution
  - Probability is long-run frequency

$P(X=0.7)$ infinite সংখ্যক বার sampling করলে 70% case এ head, 30% এ tail

θ estimation এ uncertainty নেই
sampling এ " আছে

sampling ঠিকমতো করা হলে যে theta পাবে সেটা ভালো estimate হবে।

- Bayesian
  - Uncertainty in estimates is quantified through the posterior distribution
  - Probability is degree of belief

data দেখার পরে belief change হতে পারে

data observe করা হয়ে গেলে θ comes from a distribution

Bayes' Rule

- Given two events A and B,

$$\mathrm{P}(A, B) = \mathrm{P}(A|B)\mathbb{P}(B) = \mathrm{P}(B|A)\mathrm{P}(A)$$

and,

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)}$$

Bayesian Approach

- Given (training) data D and a parameter θ $(random\ variable)$,
- Likelihood, $\mathrm{P}(D|\theta)$
- Prior, $\mathrm{P}(\theta)$
- Evidence, $\mathrm{P}(D)$
- Posterior,

$$\mathrm{P}(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Posterior $\propto$ Likelihood $\times$ Prior

frequentist approach → θ এর একটা value পাই। prediction function বসাতে পারি।

Bayesian এ → θ'র distribution পাই। সেটার উপর expectation বের করা লাগবে।

Bayesian Approach

- Predictive distribution

$$\mathrm{P}(y^*|x^*, D) = \int \underbrace{P(y^*|x^*, \theta)}_{\text{expected value of probability under}} P(\theta|D)\, d\theta$$

distribution

- Integral over all values of θ using the posterior probability estimated earlier.

Coin Flipping

- Estimate the bias θ of a coin, i.e., the probability of heads, where $\theta \in [0, 1]$
- N flips of a coin has yielded k heads.

Coin Flipping

- Frequentist Approach
  - Coin flips are independent and identically distributed Bernoulli trials with unknown parameter θ
  - θ is fixed but unknown

  - The likelihood of observing k heads out of n flips is:

$$L(\theta) = P(\text{data}|\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$$

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Coin Flipping

- Frequentist Approach
  - We pick θ to maximize the likelihood
  - If $\hat{\theta}_{MLE}$ is the value of θ that maximizes the likelihood then,

$$\hat{\theta}_{MLE} = \arg\max_{\theta} L(\theta)$$

  - Since $\binom{n}{k}$ doesn't depend on θ we can maximize the log likelihood as follows,

$$\log L(\theta) = k \log \theta + (n-k)\log(1-\theta)$$

Coin Flipping

- Frequentist Approach
  - Setting derivative w.r.t. θ to zero, we obtain,

$$\hat{\theta}_{MLE} = \frac{k}{n}$$

  - MLE simply yields the empirical proportion of heads observed.
  - No priors about the bias of the coin is incorporated
  - Often confidence intervals are reported or hypothesis tests are performed

Coin Flipping

- Bayesian Approach
  - The coin flips are i.i.d. Bernoulli trials with unknown bias θ.
  - Before observing data, we model θ as a **random variable** with a prior distribution reflecting our belief about θ.

  - What will be our prior distribution?
    - For ease of analysis, we shall pick a conjugate prior

এমন কিছু prior distribution choose করি যাদের জন্য
posterior distribution জানা আছে

Conjugate Prior

- A prior distribution
  - When combined with a specific **likelihood function**, yields a **posterior distribution of the same family** as the prior.

Likelihood (Data Model)	**Parameter**	**Conjugate Prior**	**Posterior Distribution**
Bernoulli / Binomial	θ	Beta(α, β)	Beta$(\alpha + k, \beta + n - k)$
Poisson	λ	Gamma(α, β)	Gamma$(\alpha + sum, \beta + n)$
Normal (known variance)	μ	Normal(μ, σ^2)	Normal (updated mean and variance)
Exponential	λ	Gamma	Gamma (updated parameters)

$$L(\theta) = P(\text{data}|\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$$

Coin Flipping

- Bayesian Approach
  - The coin flips are i.i.d. Bernoulli trials with unknown bias θ.
  - Before observing data, we model θ as a **random variable** with a prior distribution reflecting our belief about θ.

  - Prior
    - A conjugate prior for Bernoulli is the **Beta distribution**

$$\theta \sim \text{Beta}(\alpha, \beta)$$

  - Likelihood
    - Same as frequentist

Coin Flipping

- Bayesian Approach
  - According to Bayes' theorem,

$$P(\theta|\text{data}) \propto P(\text{data}|\theta)P(\theta)$$

  - Since Beta prior distribution was used,

$$\theta|\text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

n, k } data থেকে পাই

Coin Flipping

- Bayesian Approach
  - We have obtained a posterior distribution of θ

$$\theta|\text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

  - To get the best estimate of θ following the posterior, we use Maximum A Posteriori (MAP) method

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} P(\theta|\text{data})$$

Coin Flipping

- Bayesian Approach
  - Make predictions about the next coin flip x_{new}

$$P(x_{\text{new}}|\text{data}) = \int P(x_{\text{new}}|\theta)P(\theta|\text{data})d\theta$$

Posterior Estimation

- Posterior is intractable for most problems

- How to do that?
  - Approximate Inference (Later!!)